REVISION 2026 | SEMESTER I | THEORY COURSE | COMPLETE HANDBOOK

Fibre Science

Interactive concept, microscopic view, burning test, chemical solubility and examination handbook for Diploma Curriculum Revision 2026 — Course 1061.

Course at a glance

Course Code	1061
Contact Hours	60
Teaching Scheme	3:0:0:0
Credits	3
CIE	40
ESE	60
Bloom Level	Apply

MODULE I

Fundamentals & Classification of Textile Fibres

Gain a deep understanding of natural textile fibers, origin-based and chemical-based classifications, physical properties, and burning behavior.

11 Lecture Hours | 10 Practical Hours | CO1

1.1 Definition and Origin-Based Classification of Textile Fibres↑

A **Textile Fibre** is the fundamental unit of any textile structure. It is characterized by having a length-to-width ratio of at least 1000:1, high flexibility, and sufficient strength to be spun into yarn or processed into a fabric. Without these properties, a fibrous material is not considered a textile fiber.

Fibres are broadly classified based on their **origin (source)**:

- **Natural Fibres:** Fibres obtained from natural sources. They are further divided into:
    - *Vegetable / Cellulosic Fibres:* Cotton, Jute, Linen, Flax, Hemp, Sisal, Coir (derived from seed, bast, leaf, or fruit).
    - *Animal / Protein Fibres:* Silk (gland secretion), Wool, Mohair, Cashmere, Alpaca (hair/follicle fibers).
    - *Mineral Fibres:* Asbestos (natural silicate mineral).
- **Man-Made / Synthetic Fibres:** Fibres manufactured by chemical synthesis or regenerated from natural polymers:
    - *Regenerated (Semi-synthetic):* Viscose Rayon, Acetate, Lyocell (cellulose-based but chemically processed).
    - *Synthetic:* Polyester (PET), Nylon (Polyamide), Acrylic, Polypropylene.
    - *Inorganic Man-made:* Glass fiber, Carbon fiber, Metallic fibers.

> **Malayalam support:** Textile Fibre □□□□□□ □□□□□□□□ □□□□□□□□ □□□□□□□□□□ □□□□□□□□□□□□□ □□□□□□□ □□□□□ □□□□□□□□ □□□□□□□. □□□□□ □□□□□□□□□□□□ □□□□□□ (length-to-width ratio 1000:1 □□□□□□□□), □□□□□, □□□□□□□□□□□ □□□□□□□□□ (flexibility) □□□□□□□□□□□□□□. □□□□□□□□□□ □□□□□□ □□□□□□□□□□□□□ Natural Fibres □□□□□□, □□□□□□□□□□□ □□□□□□□□□□□□□□□□□□ Man-made/Synthetic Fibres □□□□□□ □□□□□□□□□□□□□.

1.2 Chemical Classification of Textile Fibres↑

While origin tells us where a fiber comes from, **Chemical Classification** groups fibers by their chemical composition and structural polymers:

Chemical Class	Primary Chemical Component	Examples	Key Characteristics
Cellulosic	Cellulose polymer, $(C_6H_{10}O_5)_n$	Cotton, Jute, Linen, Flax, Hemp	High moisture regain, combustible, sensitive to mineral acids, resistant to dilute alkalis.
Protein	Polypeptides / Amino acids	Silk (Fibroin), Wool (Keratin)	Highly hygroscopic, self-extinguishing, sensitive to alkalis/bleach, moderate acid resistance.
Polyamide	Amide linkages ($-CO-NH-$)	Nylon 6, Nylon 6,6	High tenacity, excellent elastic recovery, thermoplastic.
Polyester	Ester linkages ($-CO-O-$)	Polyethylene Terephthalate (PET)	Highly crystalline, hydrophobic, wrinkle-resistant.
Polyacrylic	Acrylonitrile units	Acrylic, Modacrylic	Wool-like feel, lightweight, weather-resistant.
Mineral	Silicates, glass, carbon	Asbestos, Glass fibre	Non-combustible, extremely high thermal resistance.

Malayalam support: [illegible] [illegible] [illegible] Chemical Classification. [illegible], [illegible] (Cotton) [illegible] (Jute) Cellulosic [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] Cellulose [illegible]. [illegible] (Silk) [illegible] (Wool) Protein [illegible] [illegible] [illegible] [illegible] [illegible] (Polypeptides) [illegible] [illegible].

1.3 Essential and Desirable Properties of Textile Fibres↑

To be successfully converted into yarns and fabrics, a fibrous material must possess certain properties. These are classified into **Essential (Primary)** and **Desirable (Secondary)** properties:

1. Essential (Primary) Properties

- **Length to Width Ratio (Aspect Ratio):** Must be at least 1000:1 so that fibers can twist around each other and hold together.
- **Tenacity (Fibre Strength):** Sufficient breaking strength (at least 1.0 to 1.5 grams per denier) to withstand spinning tension.
- **Cohesiveness (Spinnability):** The ability of fibers to adhere/cohere to one another when twisted. Surface friction or crimp provides this.
- **Pliability (Flexibility):** Fibers must be soft and pliable enough to bend easily without breaking during twisting and knitting/weaving.
- **Uniformity:** Moderate consistency in physical dimensions (length and fineness) is required to produce uniform, defect-free yarns.

2. Desirable (Secondary) Properties

- **Moisture Regain / Absorption:** High absorption increases comfort, reduces static charge buildup, and facilitates dyeing.
- **Resiliency and Elasticity:** Ability to recover from deformation (wrinkling or stretching). Wool has high resiliency; cotton has low.
- **Luster:** The light-reflection property that gives a fiber its shine or matte look (e.g., Silk has highly prized triangular cross-sectional luster).
- **Chemical and Environmental Resistance:** Ability to withstand exposure to acids, alkalis, sunlight, heat, and microorganisms.
- **Flammability:** Behavior towards heat and flame (safety requirement).

> **Malayalam support:** □□□□ □□□□□□□□□□ □□□□□□□□□□□□□ □□□□□□□□□□□ □□□ □□□□□□□□□ □□□□□□□□□ (Essential Properties) □□□ □□□□ □□□□□□□□□ (Desirable Properties) □□□□□□□□□□□□□□□. □□□□□□□□□□□□□□□□□□ □□□□□□□□□□: □□□□□□□□□□□ □□□□□□ □□□□□□□ □□□□□□□□□□ □□□□□□□, □□□ (strength), □□□□□□ (flexibility), □□□□□□□ □□□□□□□□□□□□□□□□□ □□□□□ (cohesiveness) □□□□□□.

1.4 Burning and Solubility Behavior of Textile Fibres ↑

Fibre identification relies heavily on physical response to flame and chemical dissolving behavior. This forms the scientific foundation for laboratory analysis.

1. Burning Test Behavior

Fibers behave distinctively when exposed to flame based on their chemical class:

Fibre Class	In Flame Response	Odor of Burning	Residue / Ash Character
Cellulosic (Cotton, Jute, Linen)	Burns quickly with yellow flame, does not melt. Continues burning after removal.	Burning paper / wood odor.	Light, fine, feather-like grey ash.
Protein (Silk, Wool)	Burns slowly with flickering flame, curls away. Self-extinguishes.	Burning hair / feathers odor.	Dark, brittle, crushable irregular bead or coal.
Synthetic (Polyester, Nylon)	Melts and shrinks away. Burns with black smoke.	Sweet chemical odor (Polyester) or celery-like odor (Nylon).	Hard, uncrushable, round dark/cream-colored plastic bead.

2. Selective Chemical Solubility

Chemical separation represents the most precise quantitative method to distinguish and isolate different fibers in blended yarns:

- **Cotton:** Dissolves in cold 70% Sulfuric Acid (H_2SO_4), but is highly resistant to hot dilute Sodium Hydroxide ($NaOH$).
- **Wool:** Dissolves completely in boiling 5% Sodium Hydroxide ($NaOH$), but is resistant to cold concentrated Sulfuric Acid.
- **Silk:** Dissolves in boiling 5% $NaOH$ and also in cold concentrated Hydrochloric Acid (HCl).
- **Polyester:** Outstanding resistance to acids and alkalis; dissolves in boiling Metacresol or boiling Nitrobenzene.

- **Nylon:** Dissolves in cold 85% Formic Acid ($HCOOH$).

> **Malayalam support:** [illegible] (Burning Test). [illegible]. [illegible]. [illegible].

1.5 Fibre Blends: Concept and Purpose ↑

In modern textiles, fibers are rarely used in isolation. A **Fibre Blend** is a yarn made by mixing two or more different types of staple fibers or filaments together during spinning. A common example is *Polycot* (Polyester + Cotton) or *Polywool*.

Purpose of Blending Fibres

Fibre blending is done systematically to achieve characteristics that no single fiber can offer:

1. **Improving Performance and Comfort:**

   Blending Polyester (strong, crease-resistant, quick-drying) with Cotton (absorbent, highly breathable, soft) creates a fabric that is comfortable to wear yet highly durable and easy to care for.

1. **Cost Optimization:** Mixing expensive premium fibers like Wool or Silk with inexpensive fibers like Polyester or Viscose makes the final garment highly affordable while retaining a luxury feel.
2. **Aesthetic Appeal and Texture:** Blending allows unique visual effects like cross-dyeing (different fibers absorbing different dyes) and creates custom textures or draping qualities.
3. **Processing Efficiency:** Blending highly static-prone synthetic fibers with natural conductive fibers reduces static electricity, making them easier to weave on automated high-speed looms.

Staple vs. Filament Fibres

- **Staple Fibres:** Short, limited-length fibers (measured in inches or millimeters). All natural fibers (except Silk) are staple fibers (e.g., Cotton is 1.5 to 5 cm long). They must be twisted together to form a yarn, giving it a fuzzy, textured appearance.
- **Filament Fibres:** Continuous, indefinite-length fibers (measured in meters or yards). Silk is the only natural filament fiber (ranging from 500 to 1500 meters). Synthetic fibers are extruded as continuous filaments which can be used directly or cut into staple lengths.

Malayalam support: [illegible] Blending [illegible]. [illegible] Poly-cot [illegible] Polyester-[illegible] Cotton-[illegible]. [illegible] (comfort) [illegible].

MODULE II

Cotton Fibre – Structure, Properties & Identification

Delve into the botanical classification, microscopic cellular morphology, chemical reactions, and properties of cotton, the king of natural plant fibers.

11 Lecture Hours | 10 Practical Hours | CO2

2.1 Classification and Botanical Varieties of Cotton ↑

Cotton is a natural seed-hair fiber. It grows in a protective capsule called a **boll** around the seeds of the cotton plant, which belongs to the genus **Gossypium** of the family **Malvaceae**.

Botanical Classification

There are four commercially cultivated species of cotton:

1. **Gossypium hirsutum:**

   Known as Upland Cotton. Native to Central America. It accounts for more than 90% of global cotton production due to its high yield and adaptability.

1. **Gossypium barbadense:** Known as Extra-Long Staple (ELS) cotton. Includes premium varieties like Egyptian, Pima, and Sea Island cottons. It yields extremely fine, long, and silky fibers of premium quality.
2. **Gossypium arboreum:** Known as Tree Cotton. Native to India and Pakistan. Highly resistant to pests and dry climates.
3. **Gossypium herbaceum:** Levant Cotton. Native to semi-arid regions of Africa and Asia. Short, coarse fibers.

Organic Cotton and BT Cotton

- **BT Cotton:** Genetically modified (GM) cotton containing genes from the bacterium *Bacillus thuringiensis*. The plant produces a toxin that selectively kills the devastating bollworm pest, reducing the need for chemical pesticides and dramatically increasing yield.
- **Organic Cotton:** Cotton grown without genetically modified seeds, synthetic pesticides, or chemical fertilizers. It promotes environmental biodiversity and soil health, fetching a premium market price.
- **Indian Hybrid Cottons:** India was the first country to develop commercial cotton hybrids (such as H4, MCU-5, and Varalaxmi) which combine high yield with long, fine staple lengths.

Major Producing Countries: China, India, United States, Brazil, and Pakistan.

> **Malayalam support:** [illegible] (boll) [illegible]. [illegible] Gossypium [illegible]. [illegible] Gossypium hirsutum (Upland cotton) [illegible]. [illegible], [illegible] BT Cotton-[illegible].

2.2 Structure and Chemical Composition of Cotton Fibre↑

The microscopic structure of cotton is unique and easily identifiable under a light microscope. As the fiber dries, the central liquid-filled core collapses, causing the fiber to twist.

1. Microscopic Morphology

- **Longitudinal View:** Appears as a flat, twisted ribbon or collapsed hollow tube. These characteristic twists are called **convolutions**. Convolutions are crucial because they increase surface friction and cohesiveness, allowing flat fibers to be spun into strong yarns without slipping.
- **Cross-Sectional View:** Appears kidney-shaped, bean-shaped, or ear-like, with a central hollow canal called the **lumen**. The lumen is the dried-up remains of the living protoplasm cell that nourished the fiber during its growth inside the boll.

2. Chemical Composition of Raw Cotton

Raw cotton is highly pure cellulose, but it contains minor natural impurities that must be removed during scouring and bleaching:

- **Cellulose:** 90% to 94% (The main structural polymer, composed of repeating glucose units).
- **Water/Moisture:** 6% to 8% (Absorbed under atmospheric conditions).
- **Proteins, Waxy matter, and Pectins:** 1% to 2% (Forms the protective outer layer of the raw fiber, making raw cotton naturally water-repellent before scouring).
- **Mineral Ash:** 1%.

> **Malayalam support:** □□□□□□□ □□□□ □□□□□□□□□□□□□□□□□□□□ □□□□□□□□□□□□ □□□□□□□ □□□□ □□□□ (convolutions) □□□□□□□□□□□□□. □□□□□□□ □□□□□□□□□□□□□ 'Lumen' □□□□□□□ □□□□□□ □□□□□. □□□□□□□ □□□ □□□□□□ 90 □□□□□□□□□□□□□□□ □□□□□□□□□□□ (Cellulose) □□□□□□□□□□□□□□□□□□□□. □□□□□□ □□□□□□□ (wax), □□□□□□□□□□ □□□□□□□□□□□.

2.3 Physical and Chemical Properties of Cotton ↑

Cotton's properties dictate its ultimate end-use in apparel, home textiles, and industrial applications.

1. Physical Properties

- **Fibre Length:** Ranges from 1.5 cm to 6 cm. Longer staple lengths make smoother, stronger, and more valuable yarns.
- **Fibre Fineness:** Highly fine. Expressed in Micronaire value (weight in micrograms per inch).
- **Tenacity (Strength):** Moderate to high (3.0 to 5.0 g/denier). Unique property: **Cotton is 10% to 20% stronger when wet**. This is because water molecules enter the amorphous regions of the cellulose polymer, forming additional hydrogen bonds that stabilize the crystalline polymer chain. This makes cotton highly washable without damage.
- **Elongation and Resiliency:** Low elastic recovery (stretches only 3% to 7% before breaking) and low resiliency. This is why cotton garments wrinkle extremely easily.
- **Moisture Regain:** Standard moisture regain is 7.5% to 8.5%. Highly absorbent and comfortable on the skin.

2. Chemical Properties and Reactions

- **Effect of Acids:** Highly sensitive. Mineral acids (such as Hydrochloric acid HCl, Sulfuric acid H_2SO_4, and Nitric acid HNO_3) hydrolyze the glycosidic linkages in the cellulose chain, causing rapid disintegration and dissolution of the fiber. Organic acids like citric or acetic acid are safe at low concentrations.
- **Effect of Alkalis:** Extremely resistant. Even boiling concentrated Sodium Hydroxide ($NaOH$) does not dissolve cotton in the absence of oxygen. When treated with cold strong $NaOH$ (15%-20%) under tension, the fibers swell, lose their convolutions, become circular, and develop a highly lustrous appearance and increased dye absorption. This chemical process is called **Mercerization**.
- **Effect of Water and Bleach:** Resistant to water. Damaged by strong oxidizing agents (bleaches) if over-exposed, which converts cellulose to oxycellulose.
- **Effect of Sunlight:** Gradual yellowing and loss of strength due to photo-oxidation.
- **Effect of Microorganisms:** Susceptible to mildew, rot, and fungi in warm, damp, humid conditions.

Malayalam support: [illegible] 10-20% [illegible]. [illegible] (Alkalis) [illegible]. [illegible] Mercerization [illegible].

MODULE III

Bast Fibres – Jute and Linen

Examine the extraction process, cellular composition, structure, and physical-chemical differences of bast fibers: Jute and Linen.

11 Lecture Hours | 10 Practical Hours | CO3

3.1 Features and Extraction of Jute Fibres↑

Jute is known as the **"Golden Fibre"** due to its golden-brown color and high commercial value. It is a bast fiber, meaning it is extracted from the phloem or inner bark of the stem of the **Corchorus** plant (species include *Corchorus capsularis* and *Corchorus olitorius*).

Jute Extraction Process (Retting and Stripping)

Extracting jute requires breaking down the pectins and gums holding the fiber bundles to the woody stem:

1 **Harvesting:**

The stalks are harvested when the flowers begin to shed.

1. **Retting:** The stems are tied into bundles and submerged in clean, slow-moving water for 10 to 20 days. Microbial action and anaerobic fermentation dissolve the sticky pectins and gums, releasing the fiber bundles from the woody core.
2. **Stripping (Extraction):** The retted stalks are taken out, and the fibers are manually stripped from the stem by beating the root ends and pulling the fiber strands.
3. **Washing and Drying:** The extracted fibers are washed in clean running water and hung on bamboo frames to dry in open sunlight for 2-3 days before baling.

Malayalam support: [illegible] Jute (Golden Fibre) [illegible]. [illegible] **Retting** [illegible]. [illegible] 10-20 [illegible]. [illegible].

3.2 Structure and Chemical Composition of Jute↑

Unlike cotton, jute is a highly lignified composite fiber, which explains its coarse texture, high stiffness, and low flexibility.

1. Microscopic Structure

- **Longitudinal View:** Appears as a bundle of smooth, overlapping polygonal cells with thick walls and irregular nodes. The fibers do not have convolutions.
- **Cross-Sectional View:** Polygonal or sharp-edged oval cells grouped in clusters, featuring a wide central hollow space (lumen) with varying diameter.

2. Chemical Composition of Jute

Jute contains a high percentage of hemicellulose and lignin, making it less pure than cotton:

- **α-Cellulose:** 58% to 63% (Provides primary tensile strength).
- **Hemicellulose:** 20% to 24% (Amorphous, highly sensitive to alkalis).
- **Lignin:** 12% to 15% (A complex natural polymer that acts as a cement, cementing the cellular fibrils together. It gives jute its stiffness, coarse feel, and brown color).
- **Fats, Wax, and Ash:** 1% to 2%.

> **Malayalam support:** [illegible] 12-15% **Lignin** [illegible]. [illegible], [illegible] polygonal [illegible].

3.3 Features and Extraction of Linen (Flax) Fibres↑

Linen is one of the oldest textile fibers known. It is extracted from the bast of the **Flax** plant, botanically classified as **Linum usitatissimum**.

Linen Extraction Process

Linen extraction is highly specialized and involves several mechanical and biological steps:

1. **Pulling:**

   Flax plants are pulled up by the roots rather

than
cut,
to
maximize
fiber
length.

1. **Retting:** The plants undergo biological retting (water retting, dew retting, or chemical retting) to break down the calcium pectate adhesive layer.
2. **Scutching:** The dried flax stems are passed through corrugated rollers to crush the brittle woody center. The stems are then beaten with a wooden paddle (scutching) to scrape away the woody fragments, releasing the long flax fibers.
3. **Hackling (Combing):** The fibers are pulled through series of metal combs (pins) to separate the short coarse fibers (called **tow**) from the long, fine, parallel fibers (called **line** fibers).

> **Malayalam support:** □□□□□□□□□ (Flax) □□□□□□□ □□□□□□□□□ Linen □□□□□□□□□□□□□. □□□□□□□ □□□□□□□□□□□□ □□□□□□□□□□□□ **Scutching** (□□□ □□□□□□□ □□□□□□□□ □□□□□□□), **Hackling** (□□□□□□ □□□□□□□ □□□□ □□□□□□□□□□□□□□) □□□□□ □□□□□□□□□□□□□□□. □□□□ □□□□□□ □□□□□□□□ 'Line' □□□□□□□ □□□□ □□□□□□□ □□□□□□□□ 'Tow' □□□□□□□ □□□□□□□□□□□□□□.

3.4 Structure, Composition, and Comparison of Jute vs. Linen ↑

Linen has a much lower lignin content than jute, giving it a bright white-cream color, high flexibility, and luxurious feel.

1. Microscopic Structure of Linen

- **Longitudinal View:** Appears as straight, cylindrical glass-like rods with characteristic swellings called **nodes** or "cross-markings" that resemble bamboo. Convolutions are absent.
- **Cross-Sectional View:** Well-defined polygons (typically 5 to 6 sided) clustered in neat bundles. The lumen is very narrow and central.

2. Chemical Composition of Linen

- **Cellulose:** 70% to 75% (Provides high strength).
- **Hemicellulose and Pectins:** 15% to 20%.
- **Lignin:** 2% to 5% (Much lower than Jute; gives strength without severe stiffness).

3. Chemical Reactions of Bast Fibres (Jute & Linen)

- **Effect of Acids:** Susceptible to hydrolysis by strong mineral acids. Both disintegrate rapidly.
- **Effect of Alkalis:** Strong alkalis dissolve the hemicellulose binding materials in Jute, causing the fiber bundles to separate and lose strength (called *splitting*). Linen is highly resistant to dilute alkalis but suffers damage under extreme conditions.

Malayalam support: [illegible] [illegible] [illegible] [illegible] [illegible] (bamboo) [illegible] [illegible] (nodes) [illegible]. [illegible] [illegible] [illegible] [illegible] (2-5%) [illegible] [illegible] [illegible]. [illegible] (Jute) [illegible] 15% [illegible] [illegible] [illegible] [illegible].

MODULE IV

Natural Protein Fibres – Silk and Wool

Study the biology, sericulture, chemical composition, properties, processing, and microscopic features of animal protein fibers: Silk and Wool.

12 Lecture Hours | 9 Practical Hours | CO4

4.1 Silk: Biology, Sericulture, and Lifecyle of Silkworm ↑

Known as the **"Queen of Textiles"**, Silk is a natural animal protein filament. It is secreted by the larvae of the silk moth during the construction of their protective cocoon.

1. Sericulture and Lifecyle

Sericulture is the commercial rearing of silkworms for the production of silk. The most widely used species is the domesticated Mulberry silkworm, **Bombyx mori**.

The life cycle of the silkworm consists of four distinct biological stages:

1 **Egg (Graine):**

The adult female moth lays hundreds of tiny eggs on mulberry leaves.

1. **Larva / Caterpillar:** The eggs hatch into larvae which feed voraciously on fresh mulberry leaves for 25 to 35 days, growing through 5 instars (molting phases). During this phase, the larva develops two large silk glands near its head.
2. **Pupa / Cocoon Formation:** When fully grown, the caterpillar spins a protective cocoon around itself by secreting a continuous double filament from its spinneret. This filament is composed of fibroin proteins held together by sericin gum. The larva transforms into a pupa inside this cocoon.
3. **Adult Moth:** If allowed to live, the pupa emerges as an adult moth in 10-14 days by secreting an alkaline fluid to dissolve the cocoon, breaking the continuous silk filament.

2. Varieties of Silk

- **Mulberry Silk:** Produced by *Bombyx mori*. Accounts for 90% of global production. Highly fine, uniform, and white-cream.
- **Wild/Non-Mulberry Silks:**
  - *Tussar Silk:* Copperish-bronze color, extracted from oak or terminalia-feeding worms.
  - *Muga Silk:* Rich golden luster, extremely durable, produced exclusively in Assam, India.
  - *Eri Silk:* Creamy white, spun-silk (since the cocoon is open-ended and the moth emerges naturally).

> **Malayalam support:** □□□□□□□□□ □□□□□□□□□□ □□□□□□□□ □□□□□□□□□ □□□□□□□□□□□□□□□□□□□□ □□□□□□□□□□ □□□□□□□□□ **Sericulture**. □□□□□□□□ □□□□□□□□□□□□□□□□ □□□□ □□□□□□□□□□□□□□□□: □□□□□ (Egg), □□□□ (Larva), □□□□□□□ (Pupa - □□□□□□ cocoon □□□□□□□□□□□□□□□□), □□□□ (Adult Moth). □□□□□□□ □□□□□□□ □□□□□□□□□□□□□□□□□□□□□□ □□□□□ □□□□□ (Mulberry silk) □□□. □□□□□□ □□□□□□□□□□□ □□□□□□□□□ □□□□□□□□ □□□□□□□ □□□□□□□□□□□ □□□□□ (Tussar), □□□ (Muga), □□□ (Eri) □□□□□ □□□□□□□□□□□□□□□□□ (Wild Silks) □□□□□.

4.2 Silk Processing: Preparation, Degumming, and Weighting ↑

Raw cocoons must undergo processing to extract the clean, continuous filament.

1. Reeling and Preparation of Raw Silk

1 **Stifling:**

The live pupae inside the cocoons are killed using dry heat or steam. This prevents the moth from emerging and breaking the filament.

1. **Boiling / Sorting:** Cocoons are sorted and boiled in hot water to soften the sticky **sericin** (silk gum).

2. **Brushing and Reeling:** The loose outer fibers are brushed to find the single continuous end of the filament. 4 to 8 filaments are combined, passed through a guide, and wound onto a reel to form a single strong thread of raw silk.

2. Chemical Composition of Raw Silk

- **Fibroin:** 70% to 75% (The insoluble structural protein core, crystalline in nature).
- **Sericin:** 20% to 25% (The gummy, amorphous protein sheath that glues the two fibroin filaments together).
- **Waxes and Inorganic Matter:** 1% to 2%.

3. Degumming and Weighting

- **Degumming:** The raw silk yarn is boiled in mild soap solution to dissolve and strip away the yellow sericin gum. This reveals the highly lustrous, soft, drapeable fibroin underneath. Silk loses about 25% of its weight during this process.
- **Weighting:** To compensate for the 25% weight loss during degumming, silk is treated with metallic salts (such as stannic chloride) which are absorbed by the fiber. Excessive weighting makes silk brittle and prone to splitting under sunlight.

4. Microscopic Appearance

- **Raw Silk:** Appears as a double filament (two fibroins) bound together by an irregular coating of sericin.
- **Degummed Silk:** Appears as smooth, regular, transparent triangular prisms that reflect light at unique angles, giving silk its characteristic luster.

> **Malayalam support:** [illegible] **Reeling**. [illegible] **Sericin** [illegible] **Degumming** [illegible]. [illegible]. [illegible] **Weighting** [illegible].

4.3 Wool: Production, Varieties, and Shearing ↑

Wool is a natural protein fiber obtained from the fleece of sheep, goats, or camelids. It belongs to the **hair-follicle** class of fibers.

1. Shearing and Varieties

- **Shearing:** The process of shaving off the fleece of the sheep in one continuous piece using mechanical clippers. This is usually done once a year in spring.
- **Varieties of Wool:**
  - *Merino Wool:* Extracted from Merino sheep. The finest, softest, and most highly prized wool with dense crimps.
  - *Coarse / Carpet Wool:* Coarser fibers used for rugs and carpets.
  - *Lamb's Wool:* Extremely soft wool shorn from sheep less than eight months old.

2. Basic Terminology

- **Crimp:** The natural wavy or spring-like waviness along the wool fiber. Crimp is crucial because it traps air pockets, providing exceptional thermal insulation and natural elasticity.
- **Felting:** The unique property of wool fibers to interlock and entangle permanently under the influence of friction, moisture, heat, and soap. This occurs due to the microscopic overlapping scales on the fiber surface.

> **Malayalam support:** □□□□□□□□□□□□□□□□□ □□□□ □□□□□□□□□ □□□□□□□□□ □□□□□□□□□□□□□□□ **Shearing** □□□□□ □□□□□□□□□□□□□□. □□□□□□□□□□□ □□□□□□□□□□□□□□□□□ □□□□□□□□□ **Crimp** □□□□□ □□□□□□□□□□□□□. □□□ □□□□□□□□ □□□□□□□□□□□□□□□□□□□□□□□□□□□ □□□□□□□ □□□□□□□□□□□□□□ □□□□□□ □□□□ (thermal insulation) □□□□□ □□□□□□□□□□□□□□. □□□□□□□□ □□□□□ □□□□□□□□□□ □□□□□□□□□□□ □□□□□□□□□□ □□□□□□□□□□ □□□□□□□ □□□□□□□ □□□□□□□□□□□□□□□□□□ **Felting** □□□□□ □□□□□□□□.

4.4 Wool Structure and Processing: Scouring, Carbonization, and Chlorination ↑

Raw wool shorn from sheep is highly contaminated with grease, sweat, dirt, and burrs. It must undergo extensive chemical processing.

1. Wool Processing Steps

1. **Scouring:**

   Raw wool is washed in hot alkaline water containing soap or synthetic detergents. This emulsifies and removes **Lanolin** (wool grease/wax) and dried sweat (suint). Lanolin is recovered as a highly valuable byproduct for cosmetics.

1. **Carbonization:** To remove cellulosic vegetable impurities (burrs, seeds, leaves) that cannot be scoured out, wool is treated with dilute sulfuric acid (H_2SO_4) followed by baking at 100°C. The acid carbonizes (burns) the cellulose into brittle charcoal, which is crushed and blown away, leaving the acid-resistant wool undamaged.
2. **Chlorination:** Treatment of wool with chlorine gas or hypochlorite solution to etch or coat the microscopic scales. This reduces wool's tendency to shrink and felt, making the wool washable in household washing machines.

2. Microscopic Structure of Wool

- **Longitudinal View:** Appears as a cylindrical tube completely covered with overlapping microscopic scales called **cuticle scales**, pointing towards the fiber tip.
- **Cross-Sectional View:** Round or slightly oval, composed of three distinct cellular layers:
  - *Cuticle:* The outer protective layer of scales.
  - *Cortex:* The main structural body containing ortho-cortex and para-cortex cells (which swell differently, producing the natural crimp).
  - *Medulla:* A hollow canal of dead cells running down the center of coarse fibers (absent in fine Merino wool).

3. Chemical Composition of Wool

Wool is composed of the protein **Keratin**. It is highly unique because it contains 3% to 4% sulfur, primarily in the form of **cystine disulfide linkages** ($-S-S-$ crosslinks) which act as internal springs, providing wool with outstanding elasticity and recovery.

> **Malayalam support:** [illegible]. [illegible] (Lanolin) [illegible] **Scouring**. [illegible] **Carbonization**. [illegible] **Chlorination** [illegible].

4.5 Physical and Chemical Properties of Silk & Wool↑

Animal fibers exhibit completely different characteristics compared to plant-based cellulosic fibers.

1. Comparison of Physical Properties

- **Fibre Form:** Silk is a continuous filament (up to 1500m); Wool is a short crimpy staple fiber (5 to 15 cm).
- **Tenacity:** Silk is extremely strong (4.0 to 5.0 g/d, comparable to steel wire of same diameter). Wool has very low tenacity (1.0 to 1.5 g/d, and drops by 20% when wet due to hydrogen bond breaking).

- **Elasticity & Resiliency:** Wool has outstanding elasticity and resiliency (recovers completely from 99% elongation). Silk has moderate elasticity.
- **Moisture Regain:** Wool has the highest moisture regain among all textile fibers (15% to 18%). Silk regain is 11%.

2. Chemical Reactions with Acids and Alkalis

- **Effect of Alkalis:** Highly sensitive. Both fibers dissolve completely in boiling 5% Sodium Hydroxide ($NaOH$). Alkalis hydrolyze the peptide bonds and break down the disulfide crosslinks in wool keratin. Cotton is completely safe in alkali, making solubility a perfect test to differentiate natural plant vs. animal fibers.
- **Effect of Acids:** Highly resistant compared to cellulose. They withstand cold dilute mineral acids. Concentrated mineral acids cause disintegration of silk and swelling/damage in wool.
- **Effect of Bleach:** Sodium Hypochlorite (chlorine bleach) rapidly degrades and dissolves wool and silk proteins, turning them yellow and brittle. Wool and silk must only be bleached using Hydrogen Peroxide (H_2O_2).

> **Malayalam support:** [illegible] (NaOH) [illegible], [illegible] ([illegible]). [illegible]. [illegible].

VISUAL LIBRARY

Fibre Morphology and Extraction Flowcharts

Examine original, responsive technical vector diagrams illustrating the longitudinal and cross-sectional views of major fibers along with the silkworm lifecycle.

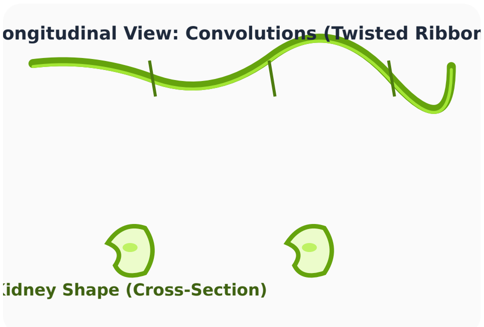

Figure 1: Cotton fibre showing characteristic twists (convolutions) longitudinally and kidney/bean shaped cross-section with central lumen.

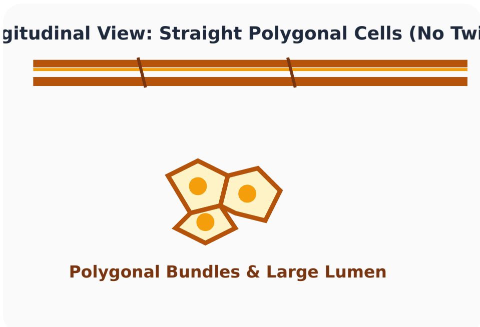

Figure 2: Jute microscopic views. Highly lignified polygonal cross-sectional cells clustered in neat bundles.

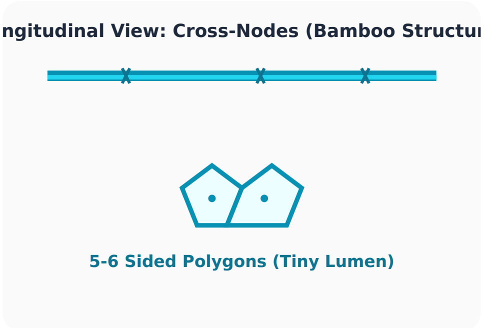

Figure 3: Linen (Flax) micro-morphology showing regular node marks (joints) and polygonal cross-section with highly constricted lumen.

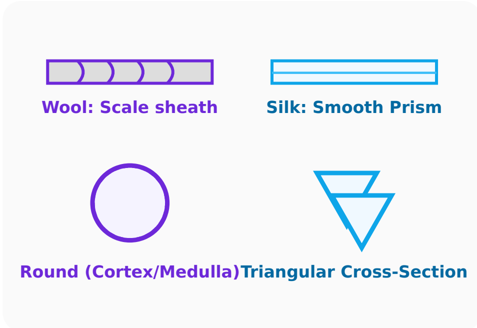

Figure 4: Microscopic structure comparison. Wool fibers possess outer protective scales. Silk filaments display smooth triangular prism cross sections giving high luster.

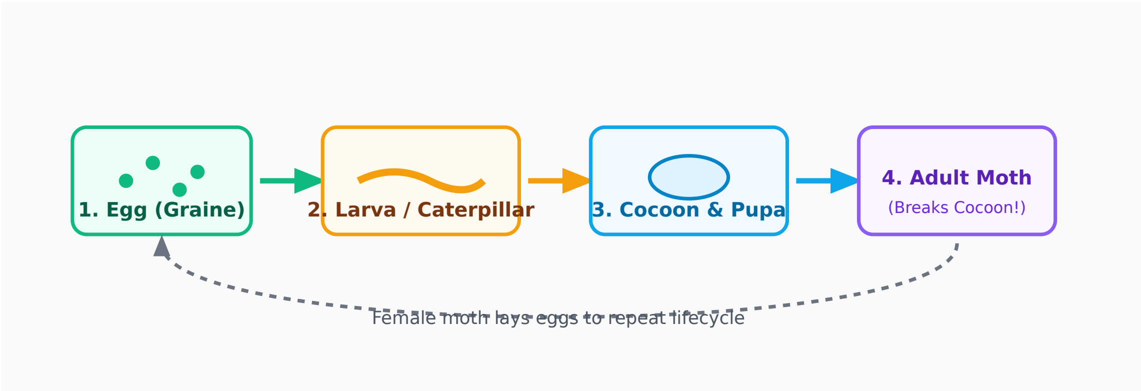

Figure 5: Biological lifecycle of Bombyx mori silkworm. Commercial sericulture harvests/stifles the cocoon at Stage 3 to preserve the continuous filament.

VIRTUAL TEXTILE LABORATORY

Interactive Fibre Identification Simulators

Hone your analytical skills by testing virtual textile fiber samples under different physical and chemical constraints.

1. Flame & Burning Test Simulator

Select a fiber sample below to simulate its burning action near a flame, inside a flame, and analyze its distinctive odor and ash residue.

Select Fibre Sample:

Simulation Status: Awaiting sample

2. Chemical Dissolution & Blend Calculator

Simulate a selective chemical dissolution test for binary fiber blends (e.g., Polyester-Cotton) to determine the percentage composition of each constituent.

Initial Dry Weight (g):

Select Solvent Reagent:

Blend Combination:

selection...

Calculated Composition: Enter parameters and run simulation.

3. Virtual Light Microscope Viewer (10x and 40x zoom)

Toggle magnification and select any natural fiber to visualize its microscopic longitudinal structure and cross-sectional morphology under the lens.

Select Fibre to Inspect:

Objective Lens Zoom:

10x Low Power 40x High Power

Select a fiber and lens magnification, then click **Focus Microscope** to generate simulated high-resolution optical morphology.

PRACTICUM MANUAL

Fibre Science Practical Lab Exercises

Step-by-step procedures, observation recording structures, chemical proportions, and setup directions for natural fiber laboratory exploration.

Experiment 1: Flame Identification of Fibres

Aim: To distinguish textile fiber samples (Cotton, Silk, Wool, Jute, Nylon, Polyester) using burning tests.

Apparatus: Bunsen burner, crucible tongs, porcelain dish, matches, sample fibers.

Procedure:

1 Hold a small cluster of the fiber sample with tongs.

1. Slowly approach the flame from the side to observe behavior *near flame*.
2. Place the sample directly into the flame to observe behavior *in flame*.
3. Remove the sample from the flame to observe if it self-extinguishes.
4. Note the burning odor and inspect the cooled ash residue left behind.

Lab Note: Cellulose □□□□□□□□□□□ □□□□□□□ □□□□□□ □□□□□□□ □□□□□□□ □□□□□□□□□. Protein □□□□□□□□□□□ □□□□□□□ □□□□□□□□□ □□□□□□ □□□□□□□□□□□□ □□□□□□□ □□□□□□□□ □□□□□□□□□. Synthetic □□□□□□□□ □□□□□□ □□□□□□□□□□□ □□□□□□□□□□□ □□□□□□□□□.

Experiment 2: Microscopic Examination (10x / 40x)

Aim: To prepare temporary dry mount slides and analyze the longitudinal surface features of Cotton, Jute, Linen, Silk, and Wool under a light microscope.

Apparatus: Light microscope, glass slides, cover slips, dissecting needles, glycerine, fibers.

Procedure:

(1) Separate a single, ultra-fine fiber strand using needles.

1. Place the fiber longitudinally on a clean glass slide.
2. Add one drop of water or glycerine to minimize refraction under high power.
3. Gently lower the cover slip to prevent air bubbles.
4. Examine under 10x magnification for fiber alignment, then switch to 40x for scale and node structures.

> **Lab Note:** □□□□□□□□□□ Convolutions-□□, □□□□□□□□□ □□□□□□□□ □□□□□□□□□□ Polygonal □□□□□□□□ bamboo-□□□□□□□□ □□□□□. □□□□□□□ □□□□□□□□ □□□□□□□□□□ □□□□□□□□□□□ (cuticle scales) □□□□□□□□□□□□□□□□ □□□□□□□□ □□□□□.

Experiment 3: Selective Solubility Tests

Aim: To differentiate textile fibers chemically using selective solubility reagents.

Reagents: 70% Sulfuric Acid (H_2SO_4), boiling 5% Sodium Hydroxide ($NaOH$), 85% Formic Acid, Acetone.

Procedure: Place small fiber snippets into test tubes. Add the reagent, agitate, and observe if dissolution happens in cold or hot conditions. Refer to the matrix below:

Fibre	Cold H2SO4 (70%)	Boiling NaOH (5%)	Cold Formic Acid (85%)
Cotton	**Soluble**	Insoluble	Insoluble
Wool	Insoluble	**Soluble**	Insoluble
Nylon	**Soluble**	Insoluble	**Soluble**

Experiment 4: Binary Blend Quantitative Separation

Aim: To separate and calculate the dry weight percentages of a Polyester-Cotton (Polycot) blended sample.

Procedure: Weigh a clean, dry blended specimen (W_1). Immerse in cold 70% H_2SO_4 with agitation for 30 minutes to dissolve the cotton component. Filter the remaining polyester fibers using a sintered glass crucible. Wash, dry completely in an oven, and weigh the residue (W_2).

```
Cotton \\% = \\frac{W_1 - W_2}{W_1} \\times 100, \\quad Polyester \\% =
\\frac{W_2}{W_1} \\times 100
```

> **Lab Note:** [illegible]. [illegible] desiccator [illegible].

FIBRE QUESTION BANK

Module-wise Expected Questions with Answers

A comprehensive compilation of highly structured short, medium, and descriptive academic questions with precise model answers.

Q1. Define a textile fibre. What is the minimum aspect ratio required?

Answer: A textile fiber is a unit of matter characterized by flexibility, fineness, and high ratio of length-to-thickness (at least 1000:1) which renders it spin-spinnable into yarn. It must possess adequate tensile strength and cohesiveness.

> **Malayalam:** [illegible] 1000 [illegible], [illegible] Textile fibre [illegible].

Q2. Explain the purpose and importance of blending fibers in the industry.

Answer: Blending is done to:
1. Combine properties of synthetic and natural fibers (e.g. blend strong crease-free Polyester with comfortable absorbent Cotton).
2. Reduce manufacturing costs of fabrics.
3. Introduce multi-color and visual textures during piece dyeing.
4. Minimize static electricity build-up during weaving.

Q3. Why is cotton 10% to 20% stronger when wet than dry?

Answer: When cotton is wet, water molecules penetrate the amorphous regions of cellulose, forming new, stable hydrogen bonds with hydroxyl groups. This relieves internal stresses, aligns the crystalline polymer chains, and spreads tensile load uniformly, increasing tenacity.

> **Malayalam:** [illegible] Cellulose [illegible] Hydrogen bonds [illegible]. [illegible].

Q4. Explain what Mercerization is and its benefits.

Answer: Mercerization is the treatment of cotton yarns or fabrics with cold 15-20% Sodium Hydroxide ($NaOH$) under tension. This causes the fibers to swell, losing convolutions and turning round, which increases luster, dye absorption, strength, and dimensional stability.

Q5. Detail the biological process of jute Retting.

Answer: Retting is a biological fermentation process. Jute stems are bundled and kept underwater for 10-20 days. Anaerobic bacteria and microorganisms secrete enzymes that consume pectins, gums, and hemicellulose holding the bark fiber bundles to the inner woody stick, releasing pure jute strands.

Q6. How can you differentiate Jute and Linen fibers under a microscope?

Answer: Under a microscope, Jute fibers appear straight with thick walls and wide variable lumen without nodes. Linen (Flax) fibers appear straight and cylindrical with characteristic nodes (bamboo-like joints) crossing the fiber surface. Linen has a much narrower lumen than Jute.

Q7. Define shearing and felting of wool.

Answer:
Shearing: The mechanical process of shaving off the entire wool fleece from live sheep using professional clippers.
Felting: The biological interlocking of wool cuticle scales when subjected to moisture, heat, friction, and alkaline soap, creating a dense, non-woven mat of fibers.

Q8. Why do boiling alkalis dissolve animal fibers but leave plant fibers intact?

Answer: Silk (fibroin) and Wool (keratin) are made of protein polypeptide chains. Peptide linkages and disulfide bridges are highly vulnerable to hydrolysis by hydroxyl ions (OH^-), breaking the protein backbone into soluble amino acids. Cellulosic plant fibers (Cotton, Linen) are highly resistant to basic hydrolysis.

PRACTICE EXAMINATION PAPER

Fibre Science Course 1061 Theory Practice Paper

Time: 2.5 Hours | Max Marks: 60 (Strict SITTTR Revision 2026 Curriculum Blueprint. Questions mapped from Module I to IV with internal choices.)

PART A (Answer all questions. Each question carries 1 mark)

1.	Define the term 'convolutions' in cotton.	**[1 Mark]**
2.	What is the primary chemical constituent of raw silk fiber?	**[1 Mark]**
3.	Identify one natural mineral fiber.	**[1 Mark]**
4.	State the botanical name of the Flax plant.	**[1 Mark]**
5.	What is the standard moisture regain percentage of dry wool?	**[1 Mark]**
6.	Define the term 'Lumen'.	**[1 Mark]**
7.	What byproduct is extracted during wool scouring?	**[1 Mark]**
8.	Identify the chemical compound used to dissolve Nylon fibers selectively in the laboratory.	**[1 Mark]**

PART B (Answer any 8 of 12 questions. Each question carries 3 marks)

9.	Distinguish between staple fibers and filament fibers with examples.	**[3 Marks]**
10.	List three essential (primary) properties of textile fibers and explain why they are critical.	**[3 Marks]**
11.	Write a short note on BT Cotton, explaining its genetic insect-resistance.	**[3 Marks]**
12.	Describe the chemical composition of raw cotton fiber, outlining the cellulosic content.	**[3 Marks]**
13.	Outline the manual process of 'stripping' jute fibers from retted stems.	**[3 Marks]**
14.	Why are mineral acids highly damaging to cotton fibers? Explain the reaction.	**[3 Marks]**
15.	Differentiate between the reeled 'line' flax fibers and coarse 'tow' flax fibers.	**[3 Marks]**
16.	Explain the method of degumming silk, detailing what protein is dissolved.	**[3 Marks]**
17.	What is 'Lanolin'? Describe its commercial recovery during raw wool processing.	**[3 Marks]**
18.	Explain how a burning test can differentiate a wool sample from an acrylic fiber.	**[3 Marks]**

PART C (Answer one full question from each module. Each question carries 7 marks)

MODULE I

19.A. Classify textile fibers extensively based on origin. Draw a complete block flow schematic tree. **[7 Marks]**

OR

19.B. Discuss the chemical solubility of cotton, wool, nylon, and polyester in H2SO4, NaOH, and Formic Acid reagents. Summarize inside a neat tabular matrix. **[7 Marks]**

MODULE II

20.A. Draw a detailed, neat cross-sectional and longitudinal sketch of a cotton fiber. Explain convolutions and lumen in detail. **[7 Marks]**

OR

20.B. Contrast the physical and chemical properties of cotton. Explain how mercerization increases the luster and dye uptake. **[7 Marks]**

MODULE III

21.A. Explain step-by-step how flax fibers are extracted from the stem of Linum usitatissimum, describing retting, scutching, and hackling. **[7 Marks]**

OR

21.B. Compare Jute and Linen fibers thoroughly with respect to chemical composition, physical attributes, microscopic appearance, and end-uses. **[7 Marks]**

MODULE IV

22.A. Outline the biological life cycle of Bombyx mori silkworm. Explain the commercial production and harvesting of silk from raw cocoons. **[7 Marks]**

OR

22.B. Detail the scouring, carbonizing, and chlorination of wool. Explain the chemical reasons why wool displays high resiliency and crimp. **[7 Marks]**

MODEL SOLUTIONS

Master Answer Key and Marking Guide

1. Convolutions: Natural twists along the longitudinal ribbon structure of cotton. They increase fiber friction and cohesiveness, enabling spinning.

2. Fibroin: The inner crystalline core protein. (Sericin is the outer sticky gum protein).

3. Asbestos: Natural silicate mineral fibers, highly resistant to fire.

4. Botanical Name: *Linum usitatissimum* (family Linaceae).

5. Moisture Regain of Wool: Standard value is 15% to 18% (highly hygroscopic).

6. Lumen: Central hollow canal running down cellulosic plant fibers, previously acting as the living protoplasm core during cell growth.

7. Lanolin: Raw wool grease/wax, widely recovered and purified for cosmetic and pharmaceutical ointments.

8. Nylon solvent: Cold 85% Formic Acid ($HCOOH$) dissolves polyamides selectively.

9. Staple vs Filament: Staple fibers are short, limited-length (e.g. Cotton, Wool); Filaments are long, continuous strands of indefinite length (e.g. Silk, Nylon).

10. Essential properties: 1. Aspect ratio > 1000:1 (spinnability), 2. Tenacity > 1 g/d (weaving strength), 3. Flexibility (pliability to bend into knitted/woven loops).

19.A/B scoring: Award marks for detailed taxonomic flowcharts (Vegetable, Animal, Mineral) and exact acid/alkali concentration matrices.

22.A/B scoring: Detail both sericin-to-fibroin ratios (25/75) for silk, and cystine disulfide sulfur bonds (Keratin) inside the cortex layers for wool.

INDEPENDENT STUDY

Suggested Student Activities for Self-Learning

Suggested active-learning curriculum assignments, industrial research tasks, and reporting formats to fulfill the 90-hour self-learning credits.

Module	Weekly Activity Description	Expected Outcome / Deliverable	Hours
Module I	Create a detailed flowchart with written explanations tracing the journey of one natural fiber from raw seed/source to a finished commercial garment.	Visual value chain report.	6 Hours
Module I	Study the agricultural characteristics, cultivation, and technical properties of Sisal leaf fibers.	Brief note on leaf fiber applications.	6 Hours
Module I	Write a critical essay comparing the ecological footprint and environmental sustainability of natural fibers vs. synthetic microplastics.	Sustainability research report.	6 Hours
Module II	Examine the technical characteristics of organic cotton, non-traditional medical cotton products, and non-woven hygiene wipes.	Comparative application notes.	6 Hours
Module III	Investigate the role and economic contribution of Jute crop export markets to the national and agricultural economy of India.	Economic analysis report.	6 Hours
Module IV	Explore the concept of ethical silk harvesting (Ahimsa Silk/Eri Silk) and organic wool rearing guidelines.	Ethical practices research paper.	6 Hours

TERMINOLOGY REFERENCE

Textile Fiber Glossary

Aspect Ratio: The ratio of fiber length to diameter. Must exceed 1000:1 to spin yarn successfully.

Bast Fibre: Cellulose fibers extracted from the phloem/bark of dicotyledonous plant stems (e.g. Jute, Flax).

Convolutions: Natural spiral twists on the longitudinal ribbon surface of collapsed cotton fibers.

Keratin: The sulfur-rich structural protein containing cystine crosslinks that forms wool and animal hair.

Fibroin: The insoluble structural protein core making up 75% of raw silk filament.

Sericin: The hot-water-soluble protein gum that sheaths and cements silk fibroin filaments.

Mercerization: Alkali modification of cotton with cold NaOH under tension to boost luster, absorption, and strength.

Retting: Microbial fermentation under water to digest gums/pectins from flax or jute stems, freeing fibers.

Lanolin: Valuable natural grease recovered from raw wool wash liquor during commercial scouring.

OFFICIAL RESOURCES

Syllabus Textbooks & Reference Assets

Official SITTTR Prescribed Textbooks

1. **Text Book of Fibre Science and Technology** — S. P. Mishra (New Age International).
2. **Textile Science** — Gohl & Valensky (Mahajan Publishers, Ahmedabad).

Reference Literature

- *Textile Science* — J. T. Marsh (Read Books).
- *Fundamentals of Fibre Science* — L. C. Sonntag, P. M. Joseph (John Wiley & Sons).
- *Handbook of Textile Fibres* — J. Gordon Cook (Woodhead Publishing).

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